

The background of the slide is a light blue gradient with several realistic water droplets of various sizes scattered across it. The droplets have highlights and shadows, giving them a three-dimensional appearance.

THE EFFECT OF CAFFEINE ON REACTION TIME AND NERVOUS SYSTEM ACTIVITY.

A SEMINAR PRESENTATION

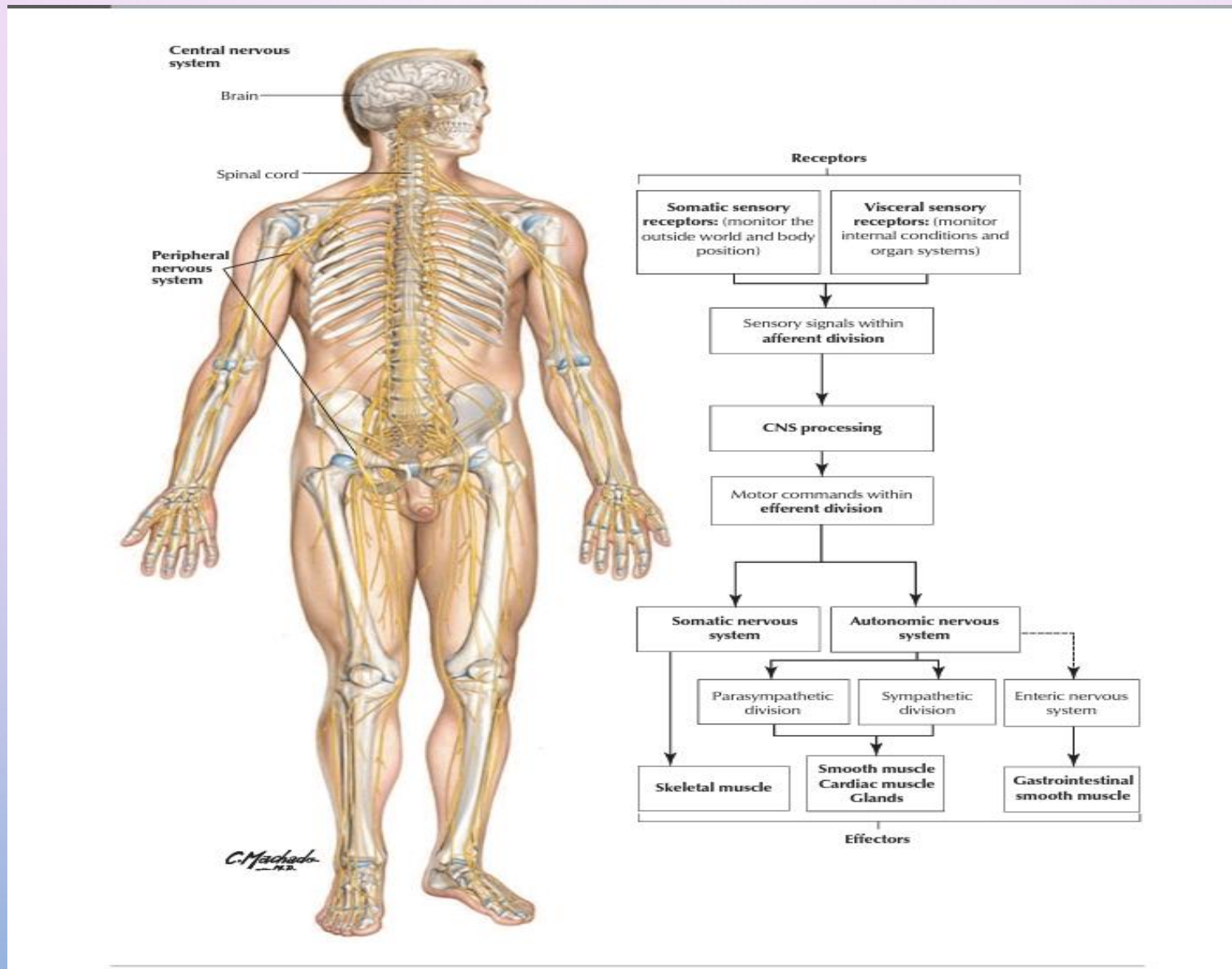
BY:

23/BM/PHS/112.

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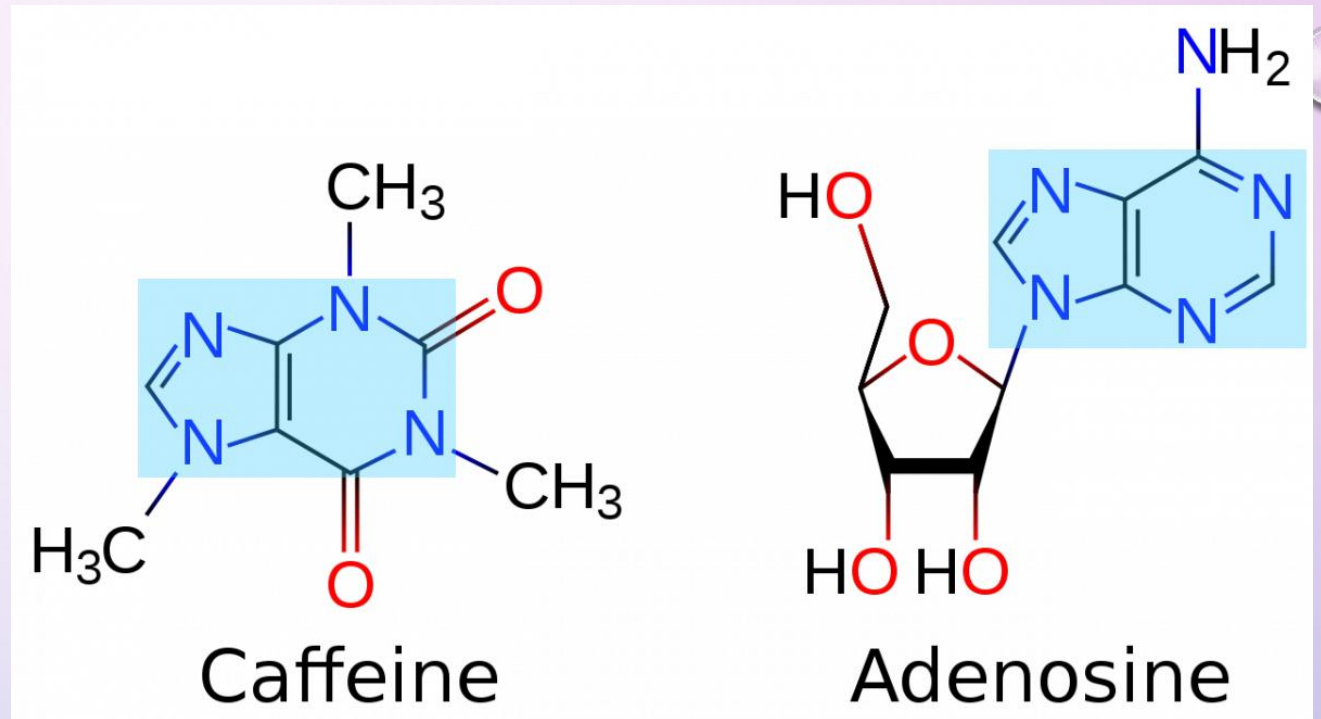
INTRODUCTION/OVERVIEW OF THE NERVOUS SYSTEM.



FUNCTIONS OF THE NERVOUS SYSTEM.

The nervous system plays a vital role in sensory perception, motor coordination, cognition, emotional regulation, and arousal. (*Guyton and hall, 2021*). These functions so listed can be influenced significantly by caffeine.

Caffeine has a chemical structure similar to adenosine, allowing it to competitively bind to adenosine receptors in place of adenosine without activating them, thus making them unavailable for adenosine to bind.



STRUCTURE OF CAFFEINE AND ADENOSINE.

MECHANISM OF CAFFEINE ACTION.

Caffeine's primary mechanism of action is antagonism of adenosine receptors in the brain. (*Fredholm et al., 1999*).

- In the brain, neurons are transmitting impulses.
- When the activity is too high, adenosine molecules stop the neurons from firing.
- Caffeine blocks adenosine receptors (A1 & A2A receptors) with its own molecule preventing adenosine from binding.
- Brain activity remains in its excited state because adenosine is unavailable to slow it down.

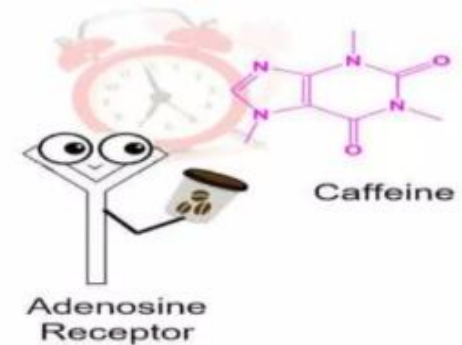
EFFECTS OF CAFFEINE.

- IT PREVENTS THE INHIBITORY EFFECTS OF ADENOSINE.

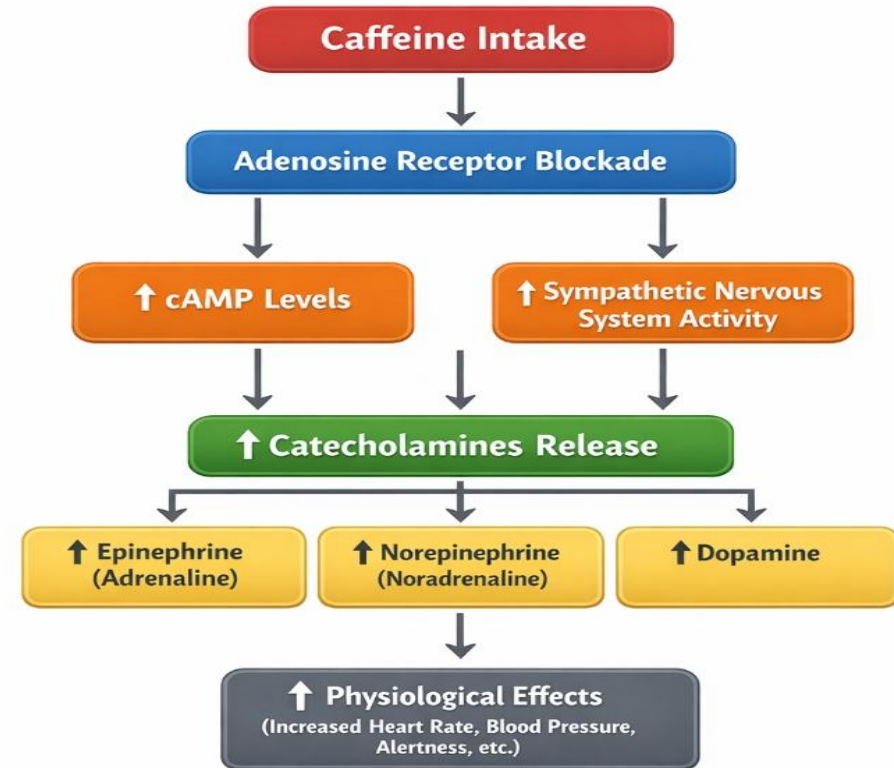
Adenosine and caffeine

The binding of Adenosine to an adenosine receptor causes the receptor to undergo a shape change which triggers a biochemical cascade. The end result is the opening of ion channels and the slowing of activity.

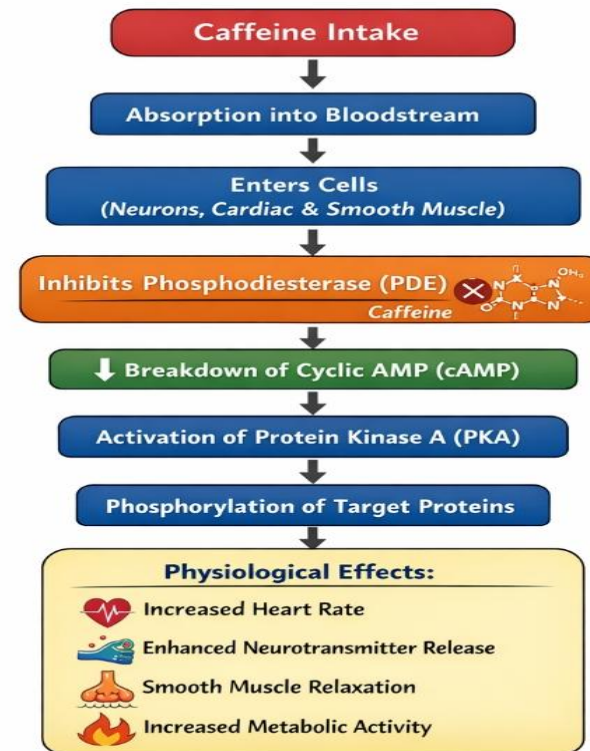
- The binding of caffeine to a adenosine receptor causes a shape change that does not initiate a biochemical cascade. Instead, neuronal activity remains the same or increase.



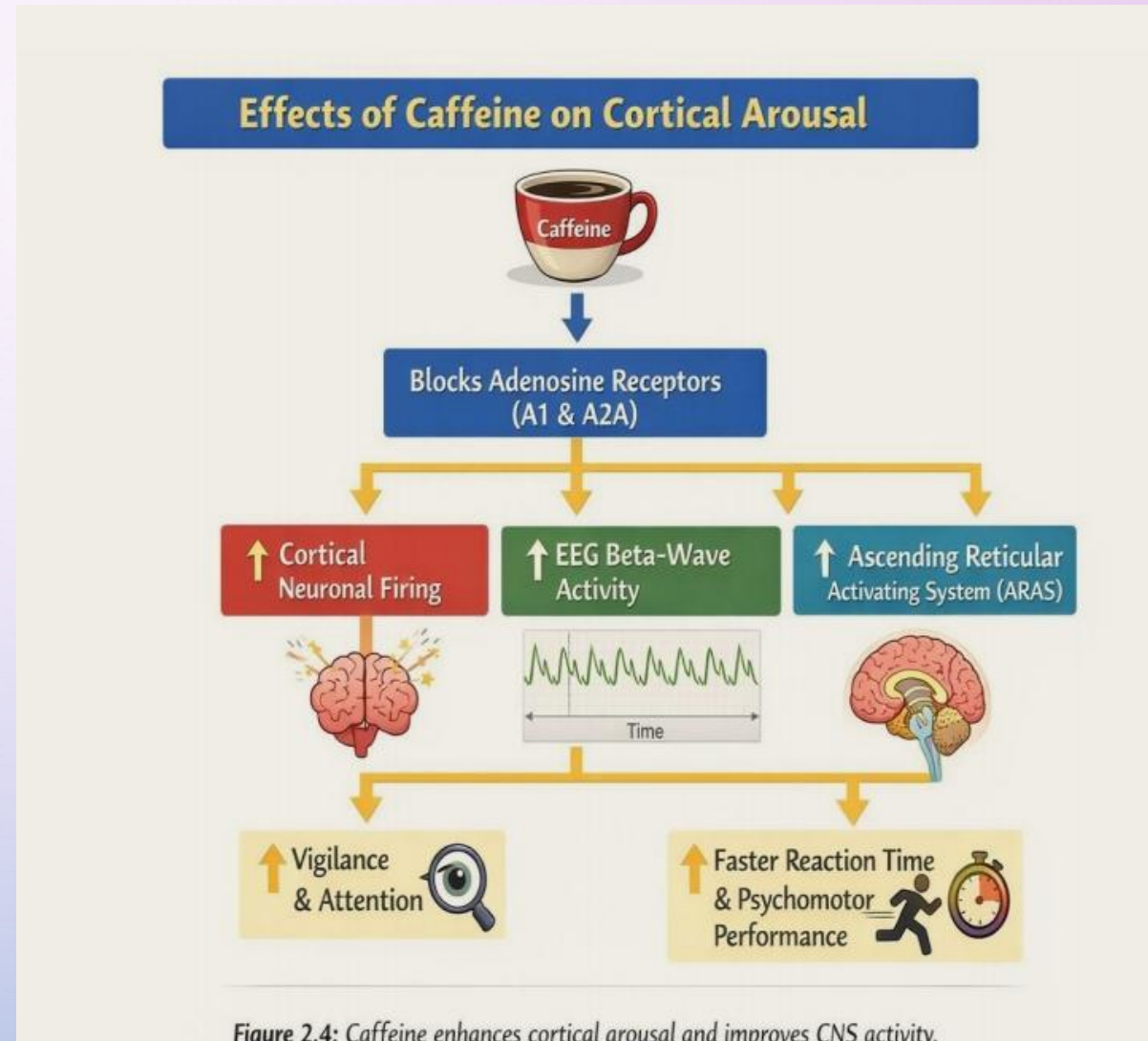
➤CAFFEINE
STIMULATES THE
RELEASE OF
CATECHOLAMINES.



➤ AT HIGH
CONCENTRATIONS,
CAFFEINE INHIBITS
PHOSPHODIESTERASE
ENZYME. (*Fredholm et
al., 1999*).

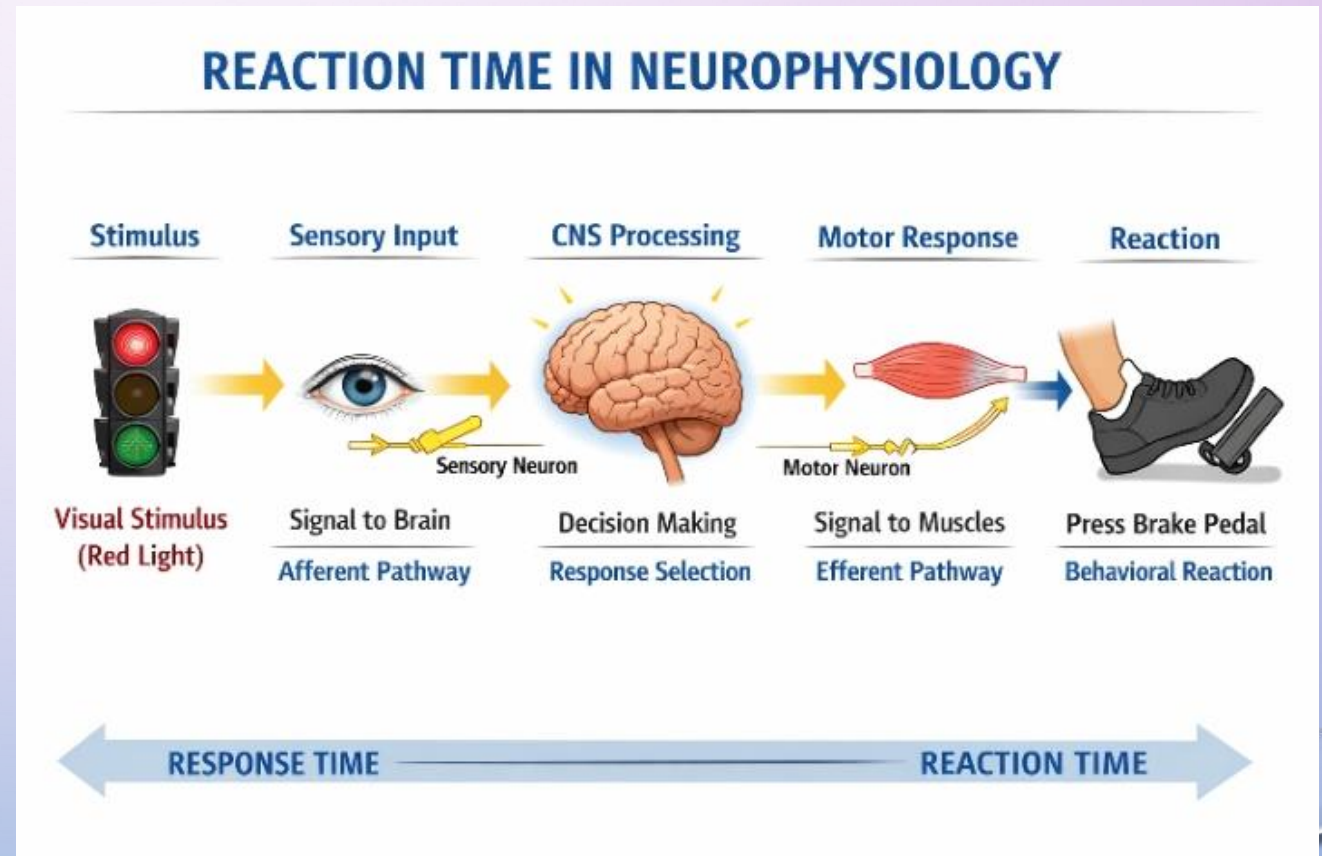


➤ Caffeine has a well documented effect on cortical arousal through its stimulatory actions on the CNS primarily via adenosine receptor antagonism and enhanced neurotransmitter release (*Fredholm et al., 1999; Nehlig, 2016*).



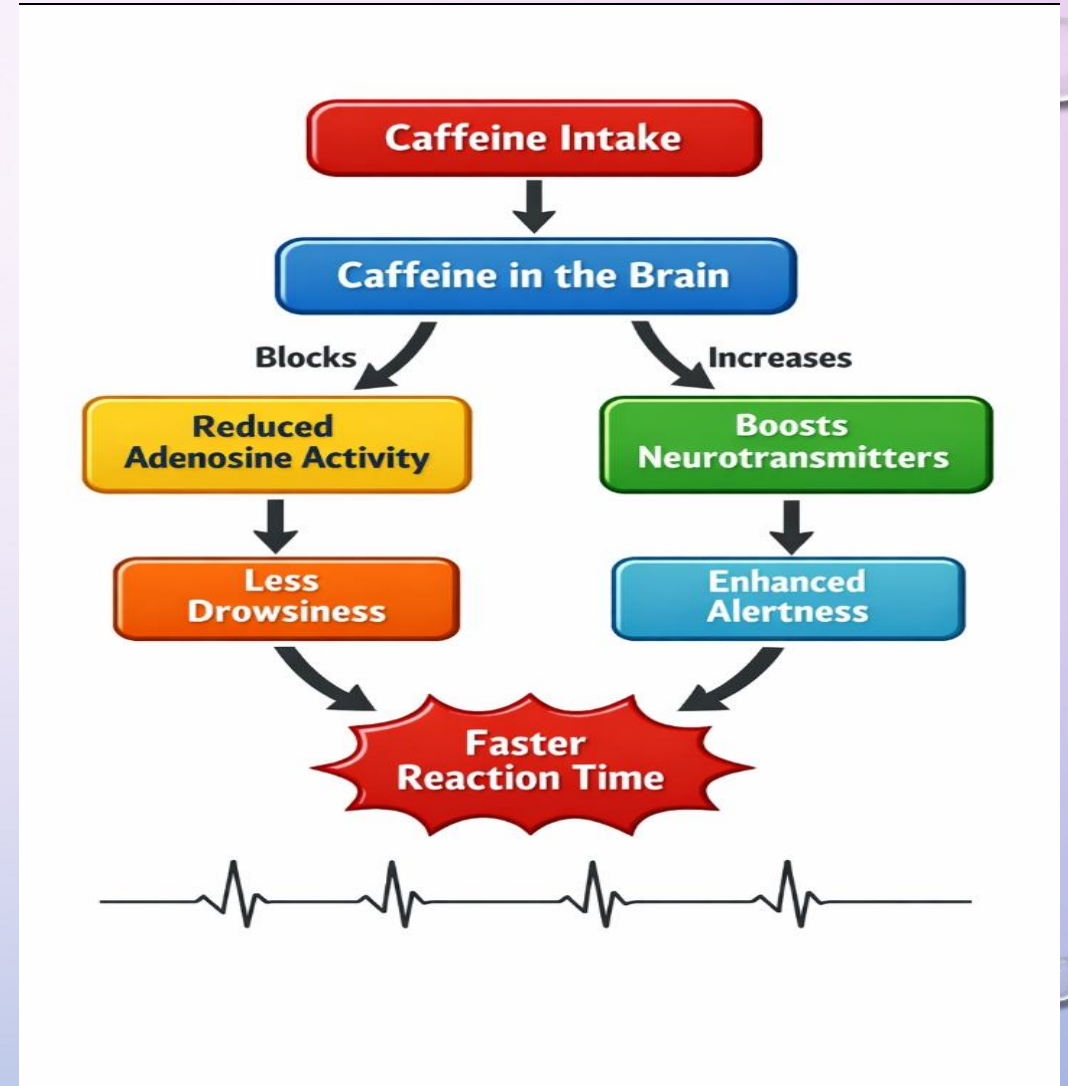
REACTION TIME.

- Is the interval between the presentation of a stimulus and the initiation of a voluntary response. It reflects the efficiency of sensory, central, and motor processes within the nervous system (Kandel *et al.*, 2021).



TO SPEED UP REACTION TIME, CAFFEINE ACTS ON:

- Adenosine receptors.
- Reticular activating system (RAS).
- α -motor neurons at neuromuscular junction (NMJ).
- Sarcoplasmic reticulum of skeletal muscles.



GUIDELINES AND RELEVANCE.

- For healthy adults, a moderate daily intake of about 200–400 mg of caffeine is generally considered safe.
- Pregnant women are advised to limit caffeine intake to about 200 mg per day.
- Caffeine should not be consumed close to bedtime, as it can interfere with sleep quality.
- In the context of this study, caffeine is relevant because of its ability to influence reaction time through increased neuronal activity and improved synaptic transmission.

CONCLUSION.

Based on the findings, caffeine has a significant effect on reaction time and central nervous system activity. It enhances alertness, improves cognitive processing, and reduces reaction time by increasing neuronal excitability and blocking inhibitory neurotransmission. The study concludes that moderate consumption of caffeine can positively influence both cognitive and motor performance.

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The background of the image features a vertical gradient from light blue at the bottom to light purple at the top. Scattered across this background are numerous water droplets of various sizes, rendered with realistic highlights and shadows to give them a three-dimensional appearance. Some droplets are clustered together, while others are isolated.

THANK YOU !!!